Lecture Scribe Tutorial 1

eg_scribe.pdf
PDF

Tutorial_1_Solution.pdf
PDF

generate a lecture scribe that can be used as exam reference material. For the content of the scribe, use the Tutorial_1_Solution pdf and for the structure of scribe use the eg_scribe pdf. Include all the definitions, notations, conditions, theorems, results and solved examples exactly given in the Tutorial_1_Solution with all the steps as mentioned in the material. Do not use any outside knowledge, do not invent or add any new explanations or examples. Also do not complete any missing proof/derivation beyond what is provided or written in the slides, just mention them. Do not use sentences or phrases like "from the slide". Maintain a formal academic tone and ensure the document is clearly organized for closed notes exam revision.

Below is the lecture scribe structured in the style of the provided example and containing only the material from **Tutorial-1: Solution**, organized formally for closed-notes examination revision.

CSE 400 – Fundamentals of Probability in Computing

Scribe: Tutorial-1 Solutions

School of Engineering and Applied Science (SEAS), Ahmedabad University

Date: February 13, 2026

---

1. Combinatorics and Multinomial Counting

Q1. Division of Dishes into Platters

Problem

20 distinct dishes are to be divided into 4 unlabeled groups of 5 dishes each.

(a) Total Number of Ways

There are 20 distinct dishes divided into 4 groups of 5 dishes each.
Since:

- Order within each group does not matter
- Order of the groups does not matter

Using the multinomial formula:

$$\text{Total ways} = \frac{20!}{(5!)^4 4!}$$

(b) Probability All Platters Have Same Cuisine

Each cuisine contains exactly 5 dishes.

The only way each platter can contain dishes from a single cuisine is if each cuisine forms its own platter.

Because platters are unlabeled, this arrangement is unique.

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}} = \frac{(5!)^4 4!}{20!}$$

---

2. Uniform Distribution and Geometric Probability

Q2. Bus Waiting Time

Arrival time is uniformly distributed over a 30-minute interval.

(a) Waiting Time < 5 Minutes

Favorable intervals:

- 7:10–7:15
- 7:25–7:30

Total favorable time = 10 minutes.

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting Time > 10 Minutes

Favorable intervals:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes.

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

---

3. Conditional Probability and De Morgan's Law

Q3. Late Arrival and Early Leaving

Let:

- L = arrives late
- E = leaves early

Given:

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

Arriving early = L^c
Leaving late = E^c

$$P(E^c) = 1 - 0.25 = 0.75$$

$$P(L \cup E) = 0.15 + 0.25 - 0.08 = 0.32$$

Using De Morgan's Law:

$$P(L^c \cap E^c) = 1 - 0.32 = 0.68$$

$$P(L^c \mid E^c) = \frac{0.68}{0.75} \approx 0.907$$

---

4. Poisson Distribution

Definition

If $X \sim \text{Poisson}(\lambda)$, then:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Q4. Typographical Errors

$$\lambda = 0.2$$

(a) $P(X = 0)$

$$P(X = 0) = e^{-0.2} = 0.8187$$

(b) $P(X \geqslant 2)$

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

$$P(1) = 0.2e^{-0.2} = 0.1637$$

$$P(X \geqslant 2) = 1 - 0.8187 - 0.1637$$

$$P(X \geqslant 2) = 0.0176 \approx 0.0175$$

Q5. Airplane Crashes

$$X \sim \text{Poisson}(3.5)$$

$$P(X = 0) = 0.0302$$

$$P(X = 1) = 0.1057$$

(a) $P(X \geqslant 2)$

$$P(X \geqslant 2) = 1 - 0.0302 - 0.1057$$

$$P(X \geqslant 2) = 0.8641$$

(b) $P(X \leqslant 1)$

$$P(X \leqslant 1) = 0.0302 + 0.1057 = 0.1359$$

Q6. Highway Accidents

$$X \sim \text{Poisson}(3)$$

$$P(0) = 0.0498$$

$$P(1) = 0.1494$$

$$P(2) = 0.2240$$

(a) $P(X \geqslant 3)$

$$P(X \geqslant 3) = 1 - 0.4232 = 0.5768$$

(b) Conditional Probability

$$P(X \geqslant 1) = 0.9502$$

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{0.5768}{0.9502} \approx 0.607$$

---

5. Gaussian Distribution and Q-Function

Q7. Gaussian Random Variable

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Comparing with Gaussian PDF:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

Gaussian relations:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

1. $P(X \leqslant 0)$

$$\Phi(\frac{3}{2}) = 1 - Q(\frac{3}{2})$$

2. $P(X > 4)$

$$Q(\frac{7}{2})$$

3. $P(|X+3| < 2)$

$$-5 < X < -1$$

$$= \Phi(1) - \Phi(-1)$$

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

4. $P(|X-2| > 1)$

$$X < 1 \text{ or } X > 3$$

$$= 1 - Q(2) + Q(3)$$

6. Binomial Distribution

Q8. Jury Decision

If defendant is innocent:

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If defendant is guilty:

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Let α be probability defendant is guilty.

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

---

7. Discrete Random Variables

Q9. Given

$$p(i) = c\frac{\lambda^i}{i!}$$

Since

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

Cumulative Distribution Function

$$F(a) = \sum_{x \leqslant a} p(x)$$

If X takes values $x_1 < x_2 < x_3 < \ldots$, then F is a step function with jumps of size $p(x_i)$.

Example provided:

$$p(1) = \frac{1}{4},\ p(2) = \frac{1}{2},\ p(3) = \frac{1}{8},\ p(4) = \frac{1}{8}$$

Corresponding stepwise CDF as specified.

---

8. System Reliability

Q10. Majority System

Number functioning is binomial (n, p).

(a) 5 vs 3 Components

5-component effective:

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5$$

3-component effective:

$$3p^2(1-p) + p^3$$

Condition reduces to:

$$3(p-1)^2(2p-1) > 0$$

$$p > \frac{1}{2}$$

(b) General Case

$$P_{2k+1} - P_{2k-1} = \binom{2k-1}{k} p^k (1-p)^k (2p-1)$$

Since binomial coefficient and powers are positive,

$$> 0 \ \Leftrightarrow\ p > \frac{1}{2}$$

End of Scribe